# SHIWAM MISHRA

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## SUMMARY

AI Engineer with 13 months of internships in agentic AI and data science. Built an LLM research agent (Deep Agents on LangGraph, FastAPI, PostgreSQL, Redis) at Tipstat, and built TabulaVibe, a live data-analysis app with 300+ automated tests. B.Pharm graduate applying AI to healthcare data.

## TECHNICAL SKILLS

**Generative & Agentic AI:** LLM APIs, Prompt Engineering, Deep Agents, LangGraph, LangChain, ReAct, Tool Calling, RAG, Embeddings, Semantic Search, ChromaDB
**Programming & Data:** Python, SQL, Pandas, NumPy, Data Cleaning, Data Validation, Feature Engineering, EDA, Statistics
**Backend & Testing:** FastAPI, REST APIs, Pydantic, Server-Sent Events (SSE), pytest, Playwright
**Machine Learning:** Scikit-learn, XGBoost, Classification, Model Evaluation, Cross-Validation, Hyperparameter Tuning, TensorFlow/Keras
**Databases & Tools:** PostgreSQL, SQLAlchemy, MySQL, Redis, Supabase, Docker, Docker Compose, Git, Vercel
**Visualization:** Power BI, Tableau, Matplotlib, Seaborn, Excel

## EXPERIENCE

**AI Engineer Intern** | Tipstat | Bengaluru (Onsite) *Mar 2026 – Sep 2026*

- Co-developed Lumen, a research agent using a **Deep Agents (LangGraph) ReAct loop** across 8 custom tools with a 3-tier search fallback (Exa, Tavily, DuckDuckGo).
- Engineered 14 endpoints across 2 **FastAPI** services, streaming agent progress via **Server-Sent Events** to a React/TypeScript UI with 4 response block types.
- Implemented **Redis** caching with per-tool expiry from 5 minutes to 24 hours, and modeled 3 core entities in **PostgreSQL** with JWT authentication.

**Data Science Intern** | Rubixe *Sep 2025 – Feb 2026*

- Delivered 4 end-to-end **machine learning** projects across healthcare, motion-sensor, gaming, and image data, from EDA to model evaluation.
- Processed datasets of up to 88,588 records with **Python and SQL**, applying joins, aggregations, imputation, and consistency checks.
- Presented trends and model results through **KPI reports and dashboards** in Power BI and Tableau to support stakeholder decisions.

## PROJECTS

**TabulaVibe AI** | FastAPI, pandas, React, TypeScript | live demo *Oct 2026*

- Built and deployed a web app that turns **CSV and Excel** uploads into a data-quality report, **KPI dashboard**, and PDF or PowerPoint reports, with Google and GitHub sign-in via Supabase.
- Developed 18 **FastAPI** endpoints for **pandas** profiling, KPI analysis, and exports, plus optional **LLM** commentary (Groq, with NVIDIA as fallback) that falls back to rules-based insights.
- Tested the app with 309 backend tests (**pytest**), 48 frontend tests (Vitest), and 10 **Playwright** end-to-end journeys on desktop and mobile.
- Cut dataset build time for a 300,000-row CSV from about 7.4 to 4.2 seconds by checking each distinct value once during type detection.

**AI Email Assistant** | FastAPI, Gmail API, Gemini, ChromaDB *Apr 2026*

- Integrated the **Gmail and Gemini APIs** in a FastAPI service running 4 LLM tasks per email (sentiment, summary, intent and urgency, draft reply), parsing **structured JSON** output.
- Embedded emails with Sentence Transformers in **ChromaDB** and retrieved the most similar past reply through semantic search as a style example for consistent tone.

**H1N1 and Seasonal Flu Vaccine Prediction** | Python, Scikit-learn, XGBoost *Jan 2026*

- Achieved a test **ROC-AUC of 0.835 (H1N1) and 0.861 (seasonal)** on 26,707 survey respondents, benchmarking 5 models with stratified cross-validation.
- Designed leak-safe **Scikit-learn pipelines** (imputation and one-hot encoding fit inside cross-validation) for features up to 50% missing and a 21% positive class.
- Identified about 53% H1N1 vaccine uptake among respondents with a doctor's recommendation versus about 14% without, using **EDA** and segmentation.

## EDUCATION

**Bachelor of Pharmacy (B.Pharm)** | Birla Institute of Technology, Mesra, Ranchi *2024*

## CERTIFICATIONS

**Certified Data Scientist**, IABAC (Mar 2026)